

OKLAHOMA CITY POST, OK, USA

WMO: 723544

Lat: 35.534N

Lon: 97.647W

Elev: 1297

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03954

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.4	18.7	1.3	6.2	18.8	6.7	8.1	24.0	29.9	43.7	27.7	47.6	10.7	0	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	100.3	73.5	97.9	73.8	94.9	74.1	77.8	91.5	76.8	90.3	75.8	89.2	12.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	131.3	84.1	72.8	127.7	83.3	72.1	124.6	82.6	42.2	91.5	41.0	90.5	40.1	89.3	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.4	25.2	23.2	6.2	104.3	4.5	3.6	3.0	107.0	0.4	109.1	-2.1	111.1	-5.4	113.8		
			5.0	80.1	4.1	2.3	2.0	81.8	-0.4	83.1	-2.7	84.4	-5.7	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.5	39.2	43.3	52.0	60.5	69.7	78.5	83.4	82.3	74.4	62.5	50.6	40.7
	DBStd	17.85	10.46	11.44	10.86	8.79	7.67	5.56	5.32	5.50	7.47	8.90	9.70	9.89
	HDD50	1162	358	244	105	18	1	0	0	0	0	12	109	315
	HDD65	3418	800	611	418	184	43	1	0	0	14	154	439	754
	CDD50	5367	23	55	168	334	610	854	1036	1003	733	399	126	26
	CDD65	2148	0	2	15	51	188	405	571	538	296	76	6	0
	CDH74	24303	2	21	127	431	1688	4443	7352	6594	2960	636	48	1
	CDH80	11718	0	4	23	99	569	2033	4029	3507	1280	170	4	0
(14) Wind	WSAvg	11.8	11.7	12.0	13.1	13.7	12.5	12.2	10.9	10.2	10.3	11.4	11.8	11.2
(15) Precipitation	PrecAvg	32.6	1.0	1.2	2.2	3.3	4.7	4.4	3.2	3.0	3.2	2.8	1.7	1.4
	PrecMax	48.7	2.8	2.6	6.5	6.8	14.2	10.6	8.0	6.9	6.9	7.8	5.7	5.0
	PrecMin	20.3	0.0	0.0	0.0	0.3	1.1	0.8	0.2	0.0	0.6	0.5	0.0	0.0
	PrecStd	7.7	0.8	0.8	1.5	1.7	2.8	2.7	2.0	1.9	1.6	1.8	1.5	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.3	77.0	83.2	87.6	93.2	99.6	104.8	104.5	100.0	88.9	79.4	70.3
		MCWB	54.0	57.6	62.3	65.4	71.5	74.0	72.6	72.5	71.5	69.3	63.5	57.8
	2%	DB	65.6	71.6	77.3	82.5	89.5	95.5	100.7	100.1	93.7	84.4	74.2	65.2
		MCWB	51.9	55.5	60.6	64.5	71.5	74.2	73.8	73.4	72.8	67.5	60.2	55.1
	5%	DB	60.9	66.1	73.1	79.1	86.1	92.5	98.5	97.4	90.7	81.1	70.2	60.8
		MCWB	49.3	54.4	59.3	63.1	70.4	73.9	74.2	73.7	72.1	65.9	58.4	50.8
	10%	DB	55.7	61.4	69.1	75.2	82.5	90.1	95.1	94.2	87.6	77.2	65.9	56.4
		MCWB	46.2	51.4	57.5	61.9	69.2	73.6	74.6	73.8	71.2	63.7	56.5	48.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	62.5	67.2	70.9	76.0	79.1	79.3	78.8	76.7	73.6	66.9	63.0
		MCDB	65.1	69.1	76.9	81.2	87.2	93.0	93.7	92.9	89.8	82.3	74.6	67.2
	2%	WB	54.9	59.7	64.3	68.4	74.0	77.4	77.8	77.3	75.1	71.3	63.2	57.3
		MCDB	61.8	66.5	73.0	77.2	84.8	90.1	91.9	91.0	87.6	79.7	70.0	61.9
	5%	WB	50.4	56.2	62.1	66.6	72.5	76.0	76.8	76.1	73.8	69.2	60.7	52.8
		MCDB	58.7	64.2	70.2	74.9	82.8	88.4	90.9	90.0	85.6	76.8	67.1	58.5
	10%	WB	46.6	51.8	59.2	64.5	70.9	74.8	75.8	75.1	72.7	66.8	58.1	48.2
		MCDB	54.9	61.2	67.0	72.5	80.6	86.6	89.8	88.9	83.8	74.5	64.9	55.6
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	20.9	21.2	21.2	19.4	18.8	19.9	19.8	19.9	20.7	20.9	19.1
		MCDBR	28.3	27.6	25.7	24.9	22.1	21.4	23.3	23.5	22.6	23.6	25.6	25.4
		MCWBR	17.3	16.2	14.1	13.1	9.7	7.8	6.5	6.9	8.1	11.6	14.7	16.3
	5% WB	MCDBR	24.8	23.3	21.0	20.1	19.2	18.6	20.0	19.3	19.1	18.6	21.0	21.4
		MCWBR	17.1	16.1	13.6	13.0	10.2	8.6	7.4	7.3	7.7	11.0	15.0	17.0
(40) Clear-Sky Solar Irradiance	taub		0.292	0.297	0.337	0.377	0.414	0.423	0.424	0.433	0.386	0.345	0.310	0.288
	taud		2.507	2.508	2.417	2.324	2.291	2.308	2.334	2.253	2.365	2.470	2.505	2.521
	Ebn at Noon		291	302	296	288	277	273	272	267	276	280	281	285
	Edh at Noon		26	29	35	40	42	41	40	43	36	30	26	24
(44) All-Sky Solar Radiation	RadAvg		885	1075	1409	1744	1888	2150	2171	1971	1632	1239	931	745
	RadStd		88	162	90	130	203	200	116	121	139	157	122	82

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153
(47) Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page